(ii) The number of students who play only Cricket = 140

(iii) The total number of students in the school = 190+110+140 =440

Aliter

(i) The number of students who play only Hockey

$n(H-C) = n(H) - n(H \cap C)$

$= 300 - 110 = 190$

(ii) The number of students who play only Cricket

$n(C-H) = n(C) - n(H \cap C)$

$= 250 - 110 = 140$

(iii) The total number of students in the school

$n(H \cup C) = n(H) + n(C) - n(H \cap C)$

$= 300+250 - 110 = 440$

WI49C

Example 1.31 In a party of 60 people, 35 had Vanilla ice cream, 30 had Chocolate ice cream. All the people had at least one ice cream. Then how many of them had,

(i) both Vanilla and Chocolate ice cream.

(ii) only Vanilla ice cream.

(iii) only Chocolate ice cream.

Solution :

Let V be the set of people who had Vanilla ice cream and C be the set of people who had Chocolate ice cream.

Then $n(V) = 35$, $n(C) = 30$, $n(V \cup C) = 60$,

Let x be the number of people who had both ice creams.

From the Venn diagram

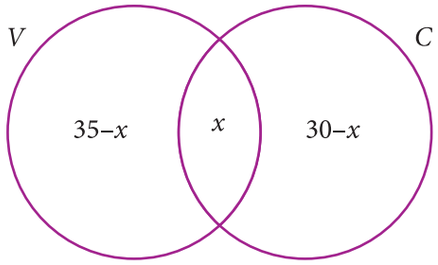


Fig. 1.35

$35 - x + x + 30 - x = 60$

$65 - x = 60$

$x = 5$

Hence 5 people had both ice creams.

(i) Number of people who had only Vanilla ice cream = $35 - x$

$= 35 - 5 = 30$

(ii) Number of people who had only Chocolate ice cream = $30 - x$

$= 30 - 5 = 25$

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